WATER SAMPLE IMAGE ORGANIZATION AND ANALYSIS

AUTHORS Sarina Talerico, Nick Darby

AFFILIATIONS Clark University Data Science Studio

ADVISORS Prof. Nathan Ahlgren, Clark University Biology Department
Prof. Li Han, Prof. Ross Potter, Clark University, Computer Science Department

MOTIVATION

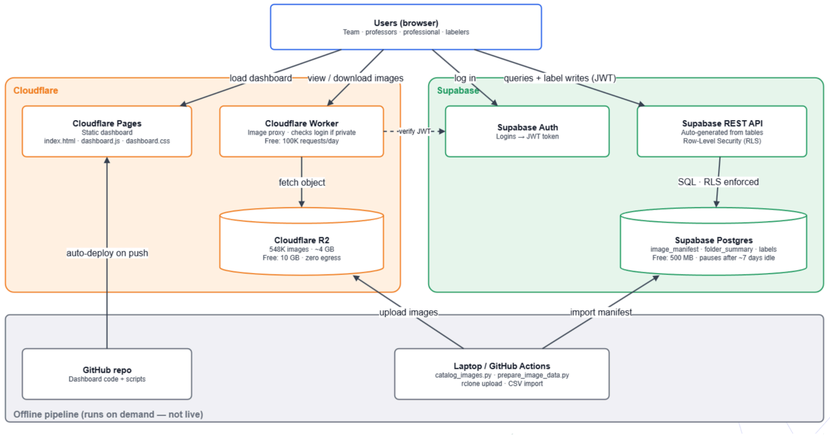
Harmful cyanobacteria blooms can make drinking water and local ponds unsafe for people and pets, and they are becoming more common as temperatures and nutrient levels rise. The City of Worcester and WCMC sample about 20 ponds and reservoirs on a regular schedule, and Prof. Ahlgren's lab studies them. Checking samples under a microscope is accurate but slow, and it takes trained experts. A FlowCam is a microscope camera that takes thousands of pictures of particles in flowing water, which makes the work much faster. It produced about 550,000 images, stored in 648 folders with inconsistent names. Nobody could easily search or compare them, and none were labeled, so a machine learning model had nothing organized to learn from.

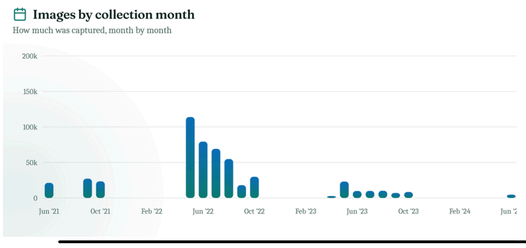
OBJECTIVE

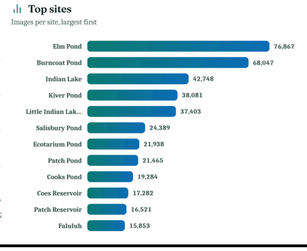
Our goal is to turn the raw FlowCam images into a clean, searchable and labelable dataset that machine learning can use. Before a model can learn to recognize cyanobacteria, the images must be organized by site, date, magnification and dilution, and a person has to label a set of them first. We set out to do three things: (1) catalog every image in a consistent format, (2) build a free, shared website where the team can browse and download the data, and (3) build a labeling tool that records who labeled each image and when. Together these steps are the foundation for faster, more consistent bloom monitoring in Worcester.

APPROACH

We worked in stages. First we learned the data by meeting with Prof. Ahlgren, who explained that -pp folders are the city's post-processed standard, what the capture-mode codes mean, and how dilution and magnification are recorded. Next we wrote a Python script to read every folder name and image and to build one catalog. Then we compared two hosting plans, one free and one at $7 per month, and chose the free one after talking with Prof. Catalin about databases. Finally we built a dashboard and a labeling tool on top of that database. At every stage we tested against the real data and fixed problems as we found them.







DESIGN ARCHITECTURE

Data Layer

Source

FlowCam raw folders: about 550,000 images
in 648 folders with inconsistent names
Prof. Ahlgren's rules: -pp = post-processed,
TR/TM and AI capture modes, dilution,
10x default magnification

Cataloging Layer

Cleaning

Python script reads folder names -> site, date,
magnification, dilution, capture mode
Uses the -pp folder when both exist
(removes 31,605 duplicate images)
Checks every image opens -> manifest +
folder summary CSV (511,765 images)

Storage Layer

Database + Images

Supabase (PostgreSQL): folders, image metadata,
accounts and labels
Cloudflare R2 + Worker: stores and serves
the images (free tier, $0 per month)

Application Layer

Dashboard + Labeling

Dashboard: browse, filter, chart, download
Labeling tool: bulk select, number-key
shortcuts, undo, progress tracking
Roles: viewer, labeler, admin (who + when saved)

Output Layer

Labeled images exported as CSV
for machine learning

YOUR PARAGRAPH TEXT

IMPLEMENTATION

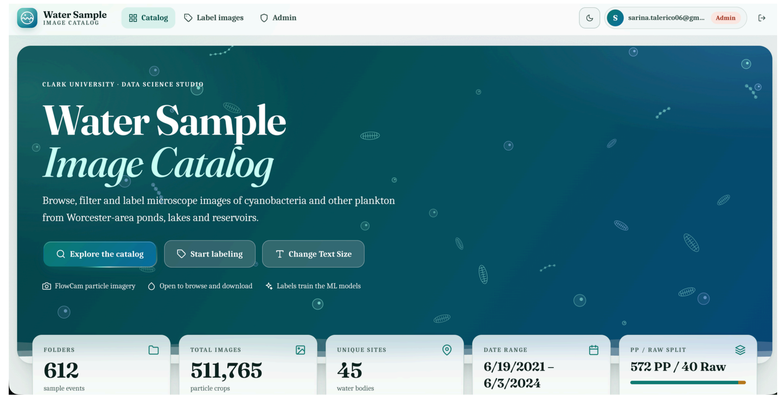
- Python with the Pillow library: cataloging and image checks
- Supabase: PostgreSQL database, sign-in, and row-level security so only approved users can label
- Cloudflare: Pages (website), Workers and R2 (image storage), $0 per month
- Vanilla JavaScript for the dashboard and labeling tool

The catalog script records one row per image (size, format, whether it opens) and one summary per folder. These go into Supabase, and the images go into Cloudflare R2. The dashboard reads the database to show filters, charts and a detail view, and the labeling page lets approved users select many images, press a number key to label them, and undo mistakes. Every label is saved with the user and the time. We also built a demo mode that works without an account, light and dark themes, and an accessible design that passed automated checks (0 violations). We also wrote a decoder that pulls images and measurements out of raw FlowCam .vbd files.

RESULTS

- 511,765 images cataloged
- 613 sample folders, 36 sampling dates
- 45 site names in the catalog (about 20 regularly sampled ponds; the rest are spelling variants and one-off samples)
- 31,605 duplicate images removed
- 0 unreadable files
- A working dashboard and labeling tool, with no hosting costs

The catalog turned scattered folders into one searchable dataset. Anyone can filter by site, date, water body and capture mode, open a folder's images, and download them. The labeling tool is built and tested.



CHALLENGES AND LIMITATIONS

- Inconsistent folder names across 648 folders, 37 folders missing magnification, and 43 raw folders that didn't match the rest
- -pp folders were not a clean filter, and some duplicated their raw versions
- Spelling variants of site names (Chauncey/Chauncy, Leesville/Leeseville) and open questions like Patch Pond vs. Patch Reservoir
- FlowCam's VisualSpreadsheet 6 can't open older datasets, so we work from exported folders
- The labeling tool isn't deployed for everyone yet, and the label categories are a placeholder until Prof. Ahlgren approves them
- We have not yet measured how accurate a model trained on our labels will be

FUTURE

This is phase 1: an organized, searchable and labelable dataset. Next we will merge the remaining site-name variants and finalize the label categories with Prof. Ahlgren. Then we will launch the labeling tool and label a varied starter set. After that comes the machine learning phase: train a small model on the labeled images, correct its mistakes, and repeat. We will also bring in the lab's existing classifier and test on samples the model hasn't seen. If it works, the city could detect blooms faster.

ACKNOWLEDGMENTS

Thank you to Prof. Li Han, Prof. Ross Potter, Prof. Nathan Ahlgren, Prof. Catalin, WCMC, and the City of Worcester.